

Impacts of Public Early Childhood Education in Chile on Maternal Fertility and Labor Market Outcomes

Short-run effects — progress update

Andrés Hojman Marigen Narea Jorge Rodríguez Pedro Carneiro Micaella Magnasco

Pontificia Universidad Católica de Chile

October 2026

Where we are

What we do

Question: does opening a public preschool nearby change mothers' labor supply and fertility?

Variation: 447 neighborhood units (UVs) open their *first* JUNJI/Integra preschool in 2015–2024 vs. 4,145 that never do

Design: staggered event study at the UV level, Sun–Abraham estimator (Baker et al., *JEL* 2026)

Data: woman-level panel linking administrative registries — RSH/FPS, SII/AFP earnings, Civil Registry births, preschool enrollment

What we find so far

Balance: in 2014, women in treated and never-treated UVs look alike; UVs differ in *size* → UV + year FE

First stage: enrollment rises $\approx 2\text{pp}$ on a 33% base ($\approx 6\%$) the year a preschool opens, persistent and robust across 3 estimators

Labor & fertility: estimates in progress

Project status

Done: data construction, treatment definition, balance, first stage

In progress: labor market and fertility event studies

Next: heterogeneity, mechanisms, writing

Research question

Question

What are the effects of the public expansion of early childhood education in Chile on mothers' labor market participation and fertility?

What we know

Mixed evidence: positive effects on maternal employment in Quebec and Argentina, close to zero in Norway

Recent Latin American evidence comes from **lotteries and experiments** (Rio de Janeiro, Nicaragua)

Less is known about a **national, decade-long public expansion** observed in administrative data

Our contribution

First study to use neighborhood-unit (UV) variation: preschools geolocated to the UV and linked to individual administrative records

Fertility as a second margin, alongside labor supply

What is a UV? *Unidad Vecinal*: the official sub-municipal territorial unit — ~6,900 UVs vs. 346 municipalities.

⇒ **Finer geography: a sharper measure of exposure and more variation to identify the effect.**

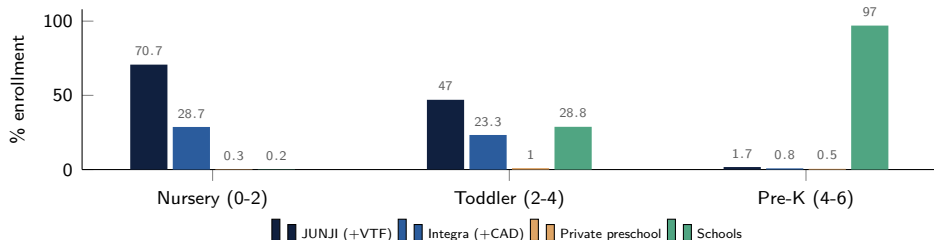
Quebec: Baker et al. (2008); Argentina: Berlinski & Galiani (2007); Norway: Havnes & Mogstad (2011); Rio: Attanasio et al. (2025); Nicaragua: Hojman & López Bóo (2022).

Public provision expanded sharply between 2015 and 2024

Mixed provision: **public** (JUNJI, Integra, municipal/SLEP) vs. **private** (subsidized/paid)

700+ new public preschools opened between 2015 and 2024

Net attendance rate (ages 0–5) rose from **50% to 55%** over the period, despite pandemic-era setbacks



Source: authors' elaboration based on Figura 5, Matrícula Oficial 2023, Centro de Estudios MINEDUC.

⇒ **Ages 0–4: JUNJI and Integra are the main providers — our focus.**

Identification: staggered-adoption event study

Design: staggered-adoption difference-in-differences with event study

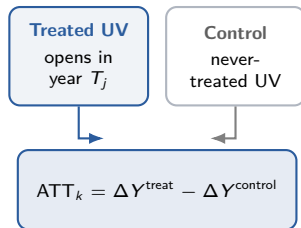
Unit of treatment: the UV, the year it opens its first public preschool

Unit of analysis: $UV \times year$ panel, built from a woman $\times$ year panel linked to each woman's UV of residence

Fixed effects: $UV + year$ (baseline); woman FE on the woman $\times$ year panel as a **robustness check**

Estimand: ATT of UV-level treatment (Baker et al., *JEL* 2026)
— an **intent-to-treat effect of access**, not individual take-up

⇒ **We compare changes in UVs that get their first preschool vs. UVs that never do — an intent-to-treat effect of access.**



A UV is treated when it opens its first preschool

Treatment definition

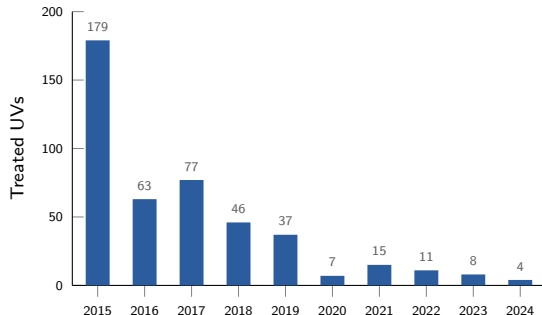
A UV is **treated** the year it opens its first public preschool (JUNJI/Integra) — goes from 0 to 1+ centers.

Group	Definition	N (UVs)	%
Excluded	Already had 1+ center in 2014 (out of design)	2,265	33.0%
Never-treated	0 in 2014, stays at 0 through 2024	4,145	60.5%
Treated	0 in 2014, opens first center 2015–2024	447	6.5%
Total universe		6,857	100%

Source: base_cobertura_cp.dta. Design sample (muestra_t1==1) = Never-treated + Treated = 4,592 UVs.

⇒ We identify the effect of the first preschool (0 → 1): 447 treated vs. 4,145 never-treated.

Treatment cohorts by year



2015 cohort is the largest: 179 UVs (40%), tied to the Bachelet preschool expansion

71% open in 2015–2017; only 45 UVs open after 2019

2014 as base year lets us identify the 2015 cohort as $0 \rightarrow 1$; without it, treated UVs drop from 447 to ~ 268

Source: base_cobertura_cp.dta. 447 treated UVs (0 centers in 2014).

⇒ **Most openings are early (2015–17): long post-treatment windows for most treated UVs.**

Empirical specification and estimator

$$Y_{ut} = \lambda_u + \lambda_t + \gamma \overline{\text{Age}}_{ut} + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - G_u = k\} + \varepsilon_{ut}$$

Y_{ut} : average outcome of women in UV u , year t (built from the woman $\times$ year panel). λ_u, λ_t : UV and year fixed effects; SE clustered by UV.

$\overline{\text{Age}}_{ut}$: average age of mothers in the UV-year (labor market specification).

β_k : effect k years from opening, relative to $k = -1$ (omitted); $k \leq -5$ binned.

Estimation strategy

Baseline TWFE as reference — but with staggered treatment, TWFE can use already-treated units as controls (Goodman-Bacon problem). Robust estimator: **Sun–Abraham** (`eventstudyinteract`), never-treated UVs as control group. Framework: Baker, Callaway, Cunningham, Goodman-Bacon & Sant’Anna (*JEL*, 2026).

Main threat: non-random placement of openings (JUNJI/Integra prioritizing UVs by care demand or socioeconomic profile). **UV FE** absorb permanent differences across UVs; **pre-period coefficients** ($k < -1$) test whether trends were already diverging.

⇒ **Main estimator: Sun–Abraham with never-treated UVs as controls; TWFE as reference.**

Data: what we have and how each sample is built

Sample	Source data	Key variables
Coverage (UV $\times$ year)	JUNJI/Integra transparency requests + UV shapefile (MDSF)	Number of centers, treatment status, opening cohort
Women panel (base)	RSH (2016–2024) + FPS (2014–2015)	UV of residence, UV population, child density
Labor market	+ Rentas (SII/AFC) + Cotizaciones (AFP)	Annual income, months worked, employment
Fertility	+ Civil Registry birth records (via RIS)	Birth in the year, children under 5
First stage (enrollment)	+ Matrícula parvularia (child-level)	Enrollment, enrollment rate

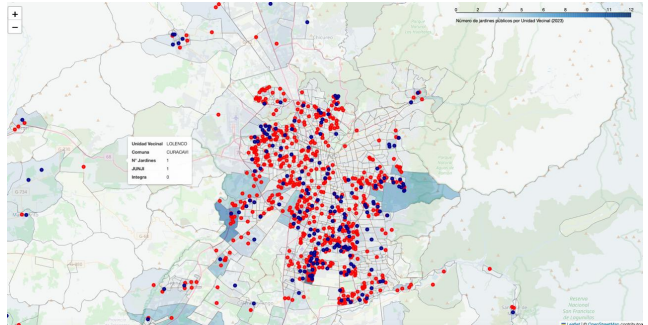
⇒ All administrative registries, linked at the individual level — no survey self-reports.

Every preschool is matched to its 2024 neighborhood unit

Each JUNJI/Integra preschool is **geolocated** and matched to the UV where it sits (spatial join)

UV boundaries **changed a lot** over time
→ we traced them and mapped everything to **2024 boundaries**

Preschools and women's UV of residence share **one common geography** across all years

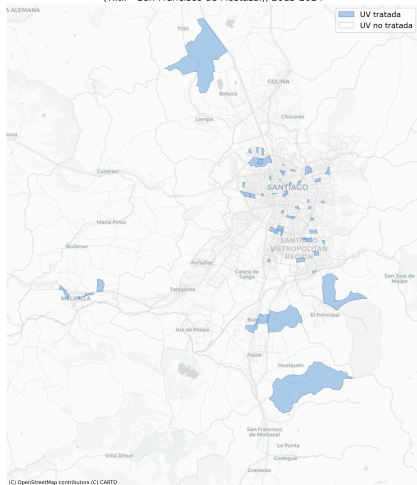


Región Metropolitana. ● JUNJI ● Integra Shading: public preschools per UV. Source: base_cobertura_cp.dta.

⇒ **One common geography: every preschool and every woman mapped to 2024 UV boundaries.**

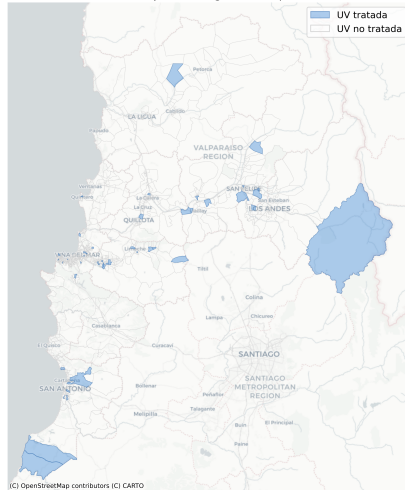
Región Metropolitana

UVs tratadas ($0 \rightarrow \geq 1$ jardín), Gran Santiago
(Tiltil - San Francisco de Mostazal), 2015-2024



Valparaíso

UVs tratadas ($0 \rightarrow \geq 1$ jardín), Región de Valparaíso, 2015-2024



Illustrative examples. Shaded: treated UVs ($0 \rightarrow 1+$ preschools), 2015–2024.

Who is in each sample, and what we measure

1. **Samples** (selected at the individual level)

Fertility: women of fertile age (18–49)

Labor market: women with a child < 5 *that specific year* — not “ever had a child”

First stage: children of treatment-sample mothers, birth to birth+4

2. **Outcomes** (averaged at $UV \times \text{year}$)

Fertility: share of women who had a child that year (Civil Registry birth records)

Labor market: average annual income, months worked, share employed, dependent vs. independent

First stage: enrollment rate

⇒ **Each outcome is measured on its relevant population, from administrative records, averaged at the $UV \times \text{year}$ level.**

Treated and never-treated UVs look alike — except in size

Labor market sample (mothers with a child <5) — covariate levels in 2014, before any opening

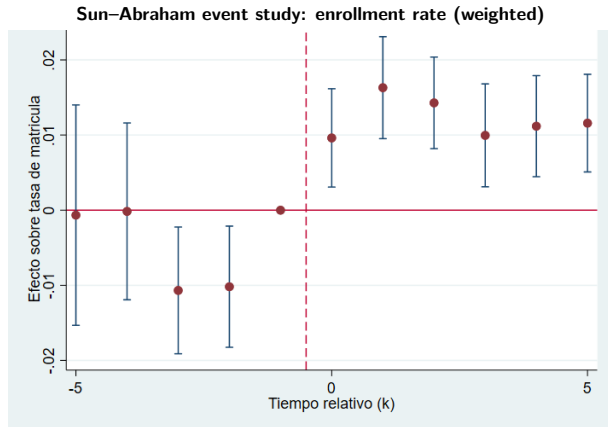
Variable	Unweighted			Weighted by UV population			Sig. ^a
	Never-tr.	Treated	Norm. diff.	Never-tr.	Treated	Norm. diff.	
Annual income (M CLP)	3.60	3.55	-0.033	3.95	3.89	-0.039	*
Months worked	8.14	8.10	-0.024	8.44	8.37	-0.058	*
Age	29.24	29.25	0.008	29.26	29.29	0.026	
N. children <5	1.11	1.11	0.045	1.11	1.11	0.049	
Hours worked	37.07	37.68	0.099	36.55	36.81	0.068	
Works (share)	0.45	0.45	0.023	0.48	0.48	-0.006	
Education (code)	4.98	5.02	0.028	5.20	5.18	-0.020	
UV population	1126.79	2011.08	0.410	3530.92	5287.26	0.340	***
Child density 0–5 UV	0.06	0.06	0.155	0.06	0.06	0.139	***
N (UVs)	3,909 / 422			3,909 / 422			

^a t-test at the woman level (* $p < 0.05$, *** $p < 0.01$), for reference only; normalized differences prioritized (concern threshold: 0.25).

The UV population gap is **expected**: the State prioritizes new preschools where need and population are larger. Same pattern in the fertility sample (App. A4).

⇒ **Selection is in *where* preschools open (UV size), not in *who* lives there → UV fixed effects.**

Enrollment rises the year a preschool opens and stays higher



Opening a preschool raises the enrollment rate of children aged 0–4 by $\approx 1\text{pp}$ vs. never-treated UVs, on a base of $\sim 33\%$.

The effect appears in the **opening year** and persists for **at least five years**.

Sun–Abraham, weighted by number of children, never-treated UVs as controls, UV and year FE, 95% CI clustered by UV.

Reference $k = -1$.

The enrollment effect is robust across three estimators

k	Unweighted TWFE	Weighted TWFE	Sun-Abraham (unweighted)
-5	-0.002	-0.015	0.006
-4	-0.009	-0.011*	0.004
-3	-0.013 [†]	-0.018***	-0.005
-2	-0.023***	-0.017***	-0.017***
0	0.020***	0.007*	0.023***
1	0.023***	0.013***	0.026***
2	0.016***	0.012***	0.019***
3	0.018***	0.007**	0.021***
4	0.017***	0.009**	0.020***
5	0.019***	0.009***	0.023***

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, [†] $p = 0.066$. Base rate $\approx 32.6\%$. $N = 48,348$ UV-year, clusters = 4,568 UVs.

Effect: $\approx 2\text{pp}$ on a $\approx 33\%$ base ($\approx 6\%$ relative), robust across specifications and growing slightly over time. Source: `primera_etapa.txt`.

Next steps

Immediate

Finish labor market estimation

Finish fertility estimation

Short term

Disaggregate $k = -2$ by opening cohort

Heterogeneity by education level (nursery/toddler vs. pre-K)

Medium term

Robustness: weighted vs. unweighted, outlier UVs

Write results section

Appendix

Appendix A1. Full identification assumptions and placebo check

Parallel trends: assessed via pre-opening event-study coefficients; should be statistically indistinguishable from zero

No anticipation: mothers do not adjust decisions before the effective opening — depends on how much advance notice openings receive (to analyze)

Exogeneity of timing: opening timing should not correlate with unobserved UV characteristics. **Main threat:** JUNJI/Integra may prioritize UVs with higher care demand or specific socioeconomic profiles

Placebo check

An analogous event study will be estimated using the **pre-treatment birth rate** of the UV as the outcome. Non-significant pre-treatment coefficients would strengthen the parallel-trends assumption of the main model.

Appendix A2. Data limitation: UV assignment in 2014–2015

FPS (2014–2015) has no reliable UV field → those two years are anchored to the woman's **2016 UV** as a proxy

Consequence: women who left the panel before 2016 are dropped from 2014–2015 — precisely the earliest pre-trend years

RIS has not delivered a better historical crosswalk despite repeated requests; only unstandardized UV name strings are available

Idea, not yet tested: fuzzy matching on UV name strings

Appendix A3. Task status

Block	Task	Status	Note
2. Data generation	Coverage base (preschools $\times$ UV, 2014–2024)	ready	Verified; private preschools (transparency) still a possible extension
	Women panel (2014–2024, incl. FPS anchor)	ready	2014–2015 anchored to 2016 UV (see A2)
	Rentas / Cotizaciones cleaning	advanced	Independent/state/FF.AA. cotizaciones still pending
	Children registry cleaning	ready	tuvo_hijo, n_hijos_menor5
	Merge panel + outcome variables	ready	By rut_inn
	Treated / non-treated definition	ready	grupo_t1, g1, muestra_t1
3. Panel analysis	First-stage estimation (enrollment)	ready	3 specifications; $k = -2$ open issue
	Balance / validity check	ready	UV-level balance table
4. Data analysis	Methodology choice	ready	Sun–Abraham within Baker et al. (JEL 2026) framework
	Treatment definition	ready	Extensive margin, $0 \rightarrow 1$
	Run regressions (labor, fertility)	in progress	Expected before Thursday
	Analyze results	in progress	First stage analyzed; rest pending
5. Mechanisms	Heterogeneity by group / level	not started	
	Public-policy implications	not started	
6. Writing	—	in progress	

Appendix A4. Balance in the fertility sample

Fertility sample (women aged 18–49) — covariate levels in 2014, before any opening

Variable	Unweighted			Weighted by UV population			Sig. ^a
	Never-tr.	Treated	Norm. diff.	Never-tr.	Treated	Norm. diff.	
Birth rate	0.057	0.058	0.031	0.06	0.060	0.051	
Age	31.90	31.83	-0.053	31.74	31.82	0.095	
Children <5 per woman	0.303	0.319	0.156	0.31	0.320	0.117	**
UV population	1111.42	2025.96	0.423	3540.63	5305.76	0.342	***
Child density 0–5 UV	0.0590	0.0631	0.176	0.06	0.064	0.148	**
Women in sample per UV	233.60	445.45	0.401	795.29	1234.46	0.347	***
N (UVs)	3,973 / 422			3,973 / 422			

^a t-test at the UV level (** $p < 0.01$, *** $p < 0.001$), for reference only; normalized differences prioritized (concern threshold: 0.25). UVs with at least one sample woman in 2014, hence more never-treated UVs than in the labor sample.

⇒ **Same conclusion as the labor sample: women look alike, UVs differ in size.**